

Bulk Orders & Operational Efficiency Analysis

Business Objective

This analysis focuses on identifying:

1. Which customers drive large-volume sales
2. Which products create bulk purchasing behavior
3. Which customers may be operationally inefficient
4. Whether wholesale or retail channels dominate bulk purchasing
5. Whether discounts contribute to large orders

The goal is to produce actionable operational insights for:

- inventory planning
 - customer strategy
 - pricing decisions
 - logistics optimization
 - sales prioritization
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Step 1 — Load & Clean the Dataset

```
import pandas as pd
import matplotlib.pyplot as plt

# Load dataset

df = pd.read_csv('Tech Sales Dataset.csv')

# Clean column names

df.columns = df.columns.str.strip()

# Convert dates

df['OrderDate'] = pd.to_datetime(df['OrderDate'])

# Convert currency fields

currency_cols = ['Price', 'Discount', 'Order Total']

for col in currency_cols:
    df[col] = (
```

```
df[col]
    .replace('[$,]', '', regex=True)
    .astype(float)
)

print(df.head())
```

Step 2 — Create Bulk Order Definitions

Businesses often define “bulk” differently.

For this analysis:

Metric	Bulk Threshold
Quantity	20+ units
Order Total	\$5,000+

Create Bulk Flags

```
# Bulk by quantity
df['bulk_quantity_order'] = df['Quantity'] >= 20

# Bulk by revenue
df['bulk_revenue_order'] = df['Order Total'] >= 5000
```

Analysis 1 — Which Customers Drive Bulk Orders?

Objective

Identify:

- strategic customers
- high-volume accounts
- revenue concentration risk

Top Customers by Revenue

```
customer_revenue = (  
    df.groupby('CustName')['Order Total']  
      .sum()  
      .sort_values(ascending=False)  
    )  
  
print(customer_revenue.head(10))
```

Visualization — Top 10 Customers by Revenue

```
customer_revenue.head(10).plot(  
    kind='bar',  
    figsize=(12,6)  
)  
  
plt.title('Top 10 Customers by Revenue')  
plt.ylabel('Revenue')  
plt.xlabel('Customer')  
plt.xticks(rotation=45)  
plt.show()
```

Operational Insight Example

A small group of customers contributes a disproportionately large share of total revenue.

Business Implications

- customer concentration risk
- need for dedicated account management
- importance of retention strategy

Analysis 2 — Which Customers Place Large Quantity Orders?

Objective

Identify customers creating operational pressure.

```
bulk_customers = (  
    df[df['bulk_quantity_order']]  
    .groupby('CustName')['Quantity']  
    .sum()  
    .sort_values(ascending=False)  
)  
  
print(bulk_customers.head(10))
```

Visualization — Bulk Customers

```
bulk_customers.head(10).plot(  
    kind='bar',  
    figsize=(12,6)  
)  
  
plt.title('Top Bulk Customers by Quantity Purchased')  
plt.ylabel('Units Purchased')  
plt.xlabel('Customer')  
plt.xticks(rotation=45)  
plt.show()
```

Business Insight Example

Wholesale-oriented customers dominate high-volume purchasing.

Operational Impact

- inventory spikes
- warehouse staffing requirements
- procurement planning

Analysis 3 — Which Products Drive Bulk Orders?

Objective

Identify products requiring:

- inventory scaling
- supplier prioritization
- operational forecasting

```
bulk_products = (  
    df[df['bulk_quantity_order']]  
    .groupby('ProdName')['Quantity']  
    .sum()  
    .sort_values(ascending=False)  
)  
  
print(bulk_products.head(10))
```

Visualization — Products Driving Bulk Sales

```
bulk_products.head(10).plot(  
    kind='bar',  
    figsize=(12,6)  
)  
  
plt.title('Top Products Driving Bulk Orders')  
plt.ylabel('Units Sold')  
plt.xlabel('Product')  
plt.xticks(rotation=45)  
plt.show()
```

Business Insight Example

Robot and drone-related products dominate large-quantity purchasing behavior.

Operational Impact

- higher inventory turnover
 - supply-chain prioritization
 - increased warehouse utilization
-

Analysis 4 — Revenue Concentration Risk

Objective

Determine whether the company depends heavily on a few customers.

```
# Total revenue

total_revenue = df['Order Total'].sum()

# Revenue by customer

customer_rev = (
    df.groupby('CustName')['Order Total']
      .sum()
      .sort_values(ascending=False)
)

# Top 10 customers contribution

top_10_pct = (
    customer_rev.head(10).sum() / total_revenue
) * 100

print(f'Top 10 customers contribute {top_10_pct:.2f}% of revenue')
```

Business Insight Example

Revenue concentration among a small customer group creates retention and forecasting risk.

Strategic Questions

- What happens if one major customer leaves?
- Should customer acquisition diversify revenue?

Analysis 5 — Retail vs Wholesale Operational Comparison

Objective

Compare operational scale across channels.

```
channel_analysis = (
    df.groupby('OrderType')
      .agg({
          'Quantity': 'mean',
          'Order Total': 'mean',
          'OrderNum': 'count'
      })
)

channel_analysis.columns = [
    'Avg Quantity',
    'Avg Revenue',
    'Order Count'
]

print(channel_analysis)
```

Visualization — Retail vs Wholesale Average Order Size

```
channel_analysis['Avg Quantity'].plot(
    kind='bar',
    figsize=(8,5)
)

plt.title('Average Quantity per Order by Channel')
plt.ylabel('Average Quantity')
plt.xlabel('Order Type')
plt.xticks(rotation=0)
plt.show()
```

Example Insight

Wholesale orders are significantly larger than retail orders.

Operational Implications

- wholesale drives inventory pressure
 - retail drives transaction volume
 - logistics strategy may need separation by channel
-

Analysis 6 — Operationally Inefficient Customers

Objective

Identify customers placing:

- many small orders
 - low-value transactions
 - operationally expensive behavior
-

Create Customer Metrics

```
customer_metrics = (
    df.groupby('CustName')
      .agg({
          'OrderNum': 'count',
          'Order Total': 'sum'
      })
)

customer_metrics.columns = [
    'Order Count',
    'Total Revenue'
]

customer_metrics['Avg Order Value'] = (
    customer_metrics['Total Revenue'] /
    customer_metrics['Order Count']
)

print(customer_metrics.head())
```

Find Potentially Inefficient Customers

Example rule:

- many orders
- low average revenue

```
inefficient_customers = customer_metrics[
    (customer_metrics['Order Count'] > 10) &
    (customer_metrics['Avg Order Value'] < 200)
]

print(inefficient_customers)
```

Visualization — Order Count vs Revenue

```
plt.figure(figsize=(10,6))

plt.scatter(
    customer_metrics['Order Count'],
    customer_metrics['Total Revenue']
)

plt.title('Customer Operational Efficiency')
plt.xlabel('Order Count')
plt.ylabel('Total Revenue')
plt.show()
```

How to Read This Chart

Area	Interpretation
Top-right	High-value customers
Bottom-right	Operationally inefficient customers
Top-left	Large one-time buyers
Bottom-left	Small customers

Example Insight

Some customers place frequent low-value orders that may increase operational costs relative to revenue generated.

Potential Business Actions

- minimum order thresholds
- shipping policy adjustments
- targeted promotions
- customer segmentation

Analysis 7 — Discount Impact on Bulk Orders

Objective

Determine whether discounts actually drive larger purchases.

```
# Discounted vs non-discounted average quantity

df['discounted'] = df['Discount'] > 0

comparison = (
    df.groupby('discounted')
      .agg({
          'Quantity': 'mean',
          'Order Total': 'mean'
      })
)

print(comparison)
```

Visualization — Discount Impact

```
comparison['Quantity'].plot(
    kind='bar',
    figsize=(8,5)
)

plt.title('Average Quantity: Discounted vs Non-Discounted')
```

```
plt.ylabel('Average Quantity')
plt.xticks(rotation=0)
plt.show()
```

Example Insight

Discounts increase average quantity only marginally, suggesting some discounts may reduce margin without significantly increasing demand.

Analysis 8 — Product Category Preference by Channel

Objective

Understand whether categories are channel-specific.

```
category_channel = pd.crosstab(
    df['ProdCategory'],
    df['OrderType'],
    values=df['Order Total'],
    aggfunc='sum'
)

print(category_channel)
```

Visualization — Category Revenue by Channel

```
category_channel.plot(
    kind='bar',
    stacked=True,
    figsize=(12,6)
)

plt.title('Product Category Revenue by Channel')
plt.ylabel('Revenue')
plt.xlabel('Product Category')
```

```
plt.xticks(rotation=45)
plt.show()
```

Example Insight

Some product categories are overwhelmingly wholesale-oriented, while others are more retail-consumer focused.

Strategic Implications

- different marketing strategies
- differentiated pricing models
- inventory allocation by channel

Executive Summary Template

After completing the analysis, your final deliverable could include findings like:

Key Findings

1. Revenue Concentration

A small group of customers contributes a large share of revenue, creating customer concentration risk.

2. Bulk Purchasing

Wholesale customers dominate bulk purchasing behavior, particularly in robotics-related product categories.

3. Operational Efficiency

Several customers place many low-value orders that may increase operational costs.

4. Product Operations

A small number of products drive most large-volume purchasing, indicating inventory prioritization opportunities.

5. Discount Strategy

Discounts may not significantly increase order quantities, suggesting margin optimization opportunities.

Recommended Business Actions

Finding	Suggested Action
Heavy revenue concentration	Improve customer retention strategy
Bulk-product demand spikes	Increase inventory forecasting
Inefficient customers	Consider minimum order policies
Weak discount effectiveness	Reevaluate discount strategy
Wholesale-driven categories	Prioritize B2B sales operations

Skills Demonstrated in This Project

This analysis demonstrates:

- data cleaning
- operational analytics
- customer segmentation
- revenue analysis
- KPI development
- visualization
- business strategy thinking
- Python analytics
- stakeholder reporting

These are highly valuable real-world data analyst skills.